

Pseudo code for merge insert

Function insert (intervals, newInterval):

result = empty list

i = 0

n = length of intervals

Step 1

Add all intervals ending before newInterval starts

while $i < n$ AND $\text{intervals}[i].\text{end} < \text{newInterval}.\text{start}$:

Append $\text{intervals}[i]$ to result

$i = i + 1$

Step 2

Merge all overlapping intervals into newInterval

while $i < n$ AND $\text{intervals}[i].\text{start} \leq \text{newInterval}.\text{end}$:

$\text{newInterval}.\text{start} = \min(\text{newInterval}.\text{start}, \text{intervals}[i].\text{start})$

$\text{newInterval}.\text{end} = \max(\text{newInterval}.\text{end}, \text{intervals}[i].\text{end})$

$i = i + 1$

Add the merged newInterval

Append newInterval to result

Step 3

Add all remaining intervals that starts after newInterval ends

while $i < n$:

Append $\text{intervals}[i]$ to result

$i = i + 1$

Return result.